

# PhaseGate: Phase-Aware CPU Retrieval Scheduling for On-Device LLMs on Unified Memory

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## Abstract

On-device assistants increasingly run GPU-based LLM inference alongside CPU tasks such as retrieval, indexing, and tool execution on the same unified-memory system. Under a saturated local-retrieval workload, four concurrent searches raise 95th-percentile (p95) decode latency by 60–61% on two M4 systems, whereas prefill latency rises by only 5.7–6.9%. A conventional scheduler applies one retrieval-concurrency limit throughout an LLM request; that limit must remain safe during sensitive decode and therefore leaves CPU capacity unused during safer prefill. On our base-M4 configuration, one concurrent search satisfies the latency limits, but two do not. PHASEGATE instead calibrates separate limits for prefill and decode, selecting four and one, respectively, on this configuration. It achieves  $2.0\times$  the latency-constrained retrieval throughput of the best tested feasible fixed policy while meeting both latency limits in every evaluation run. A phase-blind control uses the same high and low limits for the same fractions of wall-clock time but switches independently of the actual LLM phase; it achieves similar raw retrieval throughput yet violates the output-token limit in every run. M2 and M2 Pro Mac minis reproduce the policy ordering, while output-length sweeps show that the advantage narrows as decode occupies more of each request.

## 1 Introduction

Compact models now power Apple Intelligence and Gemini Nano on mobile devices [5, 6]. Yet an assistant also performs retrieval, contextual-memory lookup, indexing, parsing, and tool execution on the CPU. Representative agentic applications spend substantial time in CPU-side tools [14], and on-device vector-search systems must coordinate interactive queries with background ingestion and index maintenance [20].

Retrieval-augmented generation and agent memory embed documents or memories, find vectors similar to a query, and add the matches to the LLM prompt [2]. We use Hierarchical Navigable Small World (HNSW), a common approximate nearest-neighbor graph implemented in FAISS [15, 16]. This representative search kernel, not a complete agent workflow, performs irregular pointer chasing instead of a dense scan.

Server accelerators stream model state from GPU-local high-bandwidth memory (HBM), separate from host DRAM used by CPU tools [8, 17]. Apple Silicon instead exposes CPU and GPU to unified memory [1], making CPU pointer chasing and GPU weight streaming compete for memory service. Prior integrated-SoC work throttles CPU traffic [3], but does not ask *when* overlap is safer.

LLM phase supplies that signal: prefill processes prompt tokens in parallel and reuses weight loads, whereas decode emits one token at a time and repeatedly streams model state. On a fanless M4 MacBook Air, four HNSW searches increase p95 time-to-first-token (TTFT), our prefill-sensitive metric, by 5.7%, but time-per-output-token (TPOT), our decode metric, by 61%; a fan-cooled base-M4 Mini preserves the direction. Section 2 defines this saturated workload and shows why CPU byte rate alone misses the risk.

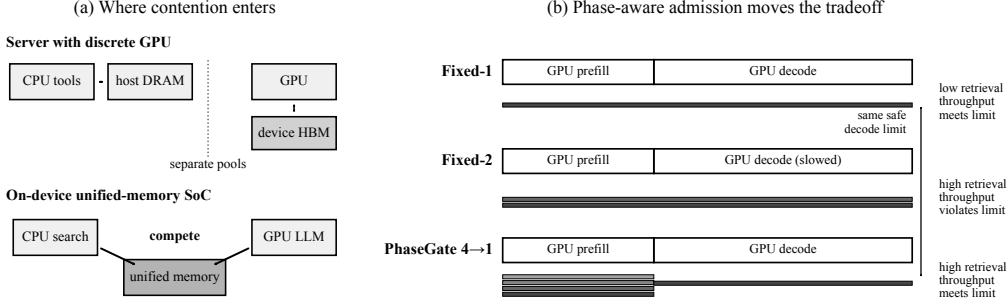


Figure 1: (a) Server GPU HBM is distinct from CPU host DRAM; an on-device CPU and GPU share one pool. (b) On the measured base-M4 configuration, Fixed-1 meets both latency limits but restricts retrieval; Fixed-2 retrieves more but violates the output-token limit. Calibrated PHASEGATE 4→1 keeps the safe decode limit while using four-way prefill concurrency. Schematic.

38 A conventional retrieval pool uses one maximum concurrency throughout an LLM request; our  
 39 Fixed policies model this phase-oblivious baseline. On the primary M4, Fixed-1 meets both latency  
 40 limits, while the faster Fixed-2 violates output-token latency. PHASEGATE instead calibrates sepa-  
 41 rate prefill and decode limits (Figure 1) and selects 4→1: four searches during prefill, but one during  
 42 decode. This pair is deployment-specific, and we compare it with a phase-blind timer that alternates  
 43 the same limits without observing LLM phase.

44 **Contributions.** (1) We establish prefill–decode asymmetry using HNSW on two M4 systems and  
 45 four access patterns on M4/M1 Airs. (2) We design PHASEGATE to preserve a safe decode limit  
 46 while using higher prefill concurrency; a time-matched phase-blind control isolates phase alignment.  
 47 (3) We evaluate the scheduler on M4, M2, and M2 Pro Minis and vary output length.

## 48 2 Observation: Phase-Asymmetric Interference

49 **Experimental Setup.** We co-run Qwen2.5-1.5B at 4-bit in MLX-LM [19] with a read-only FAISS  
 50 HNSW index (100,000 384-dimensional vectors; graph connectivity  $M = 32$ ); LLM requests use  
 51 2,048 input and 128 output tokens. Each worker handles one single-threaded query, so limit four  
 52 permits at most four independent searches. We keep the query queue nonempty to measure the  
 53 maximum sustainable retrieval throughput under an LLM-latency constraint. This saturated capacity  
 54 experiment represents waiting batched or cross-session searches, not a typical single-user trace or  
 55 prerequisite retrieval that finishes before its own request. Under intermittent arrivals, the queue is  
 56 often empty and both contention and PHASEGATE’s benefit shrink.

57 **Decode Is Much More Sensitive.** On the M4 Mini, we first measure LLM latency without re-  
 58 trieval, then allow one, two, or four concurrent HNSW searches. Each search setting uses 300  
 59 LLM requests and three repetitions; the one-, two-, and four-search settings run in randomized order.  
 60 TTFT captures the prompt-processing delay, while TPOT captures the pace of output-token gener-  
 61 ation. With two concurrent searches on the M4 Mini, p95 TTFT increases by only 1.3%, whereas  
 62 p95 TPOT increases by 31%. With four, Figure 2a shows 5.7% versus 61% on the Air and 6.9%  
 63 versus 60% on the Mini; the Mini shows greater decode sensitivity in every repetition. This pattern  
 64 is consistent with prefill amortizing weight reads while decode has little reuse.

65 **Access Pattern Matters Too.** We compare HNSW pointer chasing with bulk STREAM/memcpy,  
 66 a contiguous sequential scan, and an address-jumping random gather. Across the three controls,  
 67 p95 decode slowdown is 27–43% on M4 and 21–43% on M1, versus only 4.0–6.0% and 0.90–3.5%  
 68 for prefill. On M4, a 2.4 GB/s random gather slows decode by 29%, more than a 51 GB/s scan;  
 69 M1 preserves the ordering (26% versus 21%). Thus access pattern and LLM phase are distinct axes:

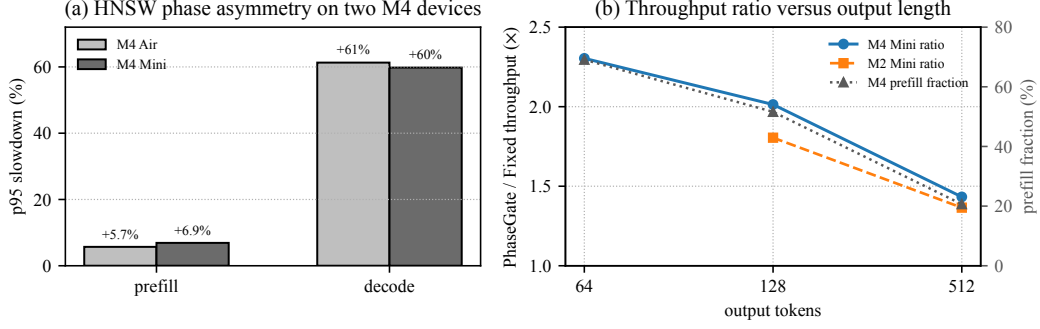


Figure 2: (a) Four concurrent HNSW searches show the same p95 asymmetry on a fanless M4 Air and fan-cooled base-M4 Mini; each Mini bar is the median of three runs with 300 LLM requests per run. (b) PhaseGate/Fixed throughput falls with output length on base-M4 (five pairs per length) and base-M2 Mini (five at 128; three at 512); the right axis shows M4 prefill fraction. Devices are analyzed separately.

byte rate alone is a poor proxy, and pattern-aware admission could further distinguish scan-like from pointer-chasing work such as HNSW [15].

### 3 PhaseGate

The goal is not merely to admit more work, but to admit it during safer prefill and restore a calibrated latency-safe limit for decode. PHASEGATE uses LLM phase, rather than CPU utilization or byte rate, to decide when a queued search may start.

**Policy and Calibration.** Let  $p$  and  $d$  be the prefill and decode concurrency limits. A phase-oblivious scheduler uses one limit throughout (e.g.,  $p = d = 1$ ); PHASEGATE permits  $p > d$ . The deployment supplies a baseline-relative budget  $B$ ;  $B = 1.25$ , for example, allows p95 TPOT and TTFT to rise by at most 25% over their no-retrieval baselines. Calibration tests candidate  $(p, d)$  pairs and selects the highest retrieval queries per second (QPS) among those that consistently meet both limits. Material changes to hardware, model, sequence lengths, retrieval, or load require recalibration.

**Runtime.** MLX-LM emits prefill and decode events; PHASEGATE admits a queued search only while the active count is below that phase’s limit. At decode, already-running non-preemptive searches finish, but new searches wait until the count falls below  $d$ . One process serves the HNSW index with a worker pool, and each query uses one FAISS thread. We request no core affinity or thread-priority (QoS) override.

**Offline Calibration, Online Phase Switching.** PHASEGATE does not adjust caps in response to latency observed during a request; it uses the known phase transition to lower the cap before decode begins. A feedback safeguard could complement it by lowering limits after unexpected token slowdown, but would require trigger and recovery thresholds.

## 4 Experimental Methodology

**Workload and Calibration Policy Sweep.** Five 300-request no-retrieval runs set M4 baselines of 12 ms TPOT and 1.8 s TTFT; every run is within 0.95% of its metric median. CPU-only tests at limits one, two, and four yield 0.98, 1.8, and 2.5 kQPS, making four the highest candidate. We test Fixed-0/1/2/4 and PhaseGate 2→0, 2→1, 4→0, 4→1, and 4→2 in three randomized 100-request runs each; eligibility requires both p95 metrics to pass every run. Across budgets 10–45% above baseline,  $B = 1.25$  is the tightest at which a retrieval-performing Fixed policy and a PhaseGate policy both remain feasible; it is a controlled comparison point, not a universal user threshold.

Table 1: Fan-cooled scheduler devices and shared setup (P/E CPU cores; UMA unified memory).

|                |                                                                        |
|----------------|------------------------------------------------------------------------|
| M4 primary     | Mac mini (Mac16,10); 4P+6E CPU, 10-core GPU, 16 GB UMA                 |
| M2 replication | Mac mini (Mac14,3); 4P+4E CPU, 10-core GPU, 16 GB UMA                  |
| M2 Pro earlier | Mac mini (Mac14,12); 6P+4E CPU, 16-core GPU, 16 GB UMA                 |
| Runtime        | macOS 26.5; MLX 0.31.2; MLX-LM 0.31.3; Qwen2.5-1.5B 4-bit              |
| Workload       | 2,048 input/128 output tokens; 100k 384-D vectors; queue kept nonempty |

Table 2: Base-M4 calibration policy sweep at  $B = 1.25$ . “Meets?” uses unrounded values and requires both p95 metrics to pass all three runs; displayed measurements use two significant digits, and all worst TTFT ratios are  $< 1.1$ . Bold policies are selected for separate evaluation.

| Fixed    | Retrieval<br>(kQPS) | Worst TPOT<br>( $\times$ baseline) | Meets?   | PhaseGate                         | Retrieval<br>(kQPS) | Worst TPOT<br>( $\times$ baseline) | Meets?   |
|----------|---------------------|------------------------------------|----------|-----------------------------------|---------------------|------------------------------------|----------|
| 0        | 0                   | 1.0                                | Y        | 2 $\rightarrow$ 0                 | 0.79                | 1.3                                | N        |
| <b>1</b> | <b>0.78</b>         | <b>1.2</b>                         | <b>Y</b> | 2 $\rightarrow$ 1                 | 1.2                 | 1.2                                | Y        |
| 2        | 1.4                 | 1.4                                | N        | 4 $\rightarrow$ 0                 | 1.2                 | 1.3                                | N        |
| 4        | 2.1                 | 1.7                                | N        | <b>4<math>\rightarrow</math>1</b> | <b>1.6</b>          | <b>1.2</b>                         | <b>Y</b> |
|          |                     |                                    |          | 4 $\rightarrow$ 2                 | 1.8                 | 1.4                                | N        |

Fixed-1 is the only tested fixed policy that both performs retrieval and is feasible; PhaseGate 2 $\rightarrow$ 1 and 4 $\rightarrow$ 1 pass, with 4 $\rightarrow$ 1 faster. We select Fixed-1 and PhaseGate 4 $\rightarrow$ 1, then evaluate them without retuning in seven randomized 250-request repetitions alongside the phase-blind control.

**Phase-Blind Control.** TimeGate uses the same limits, spends the same 51% of time at four, and switches equally often, but follows a timer rather than the LLM phase. It asks whether matching how much time is spent at each limit can reproduce the benefit without knowing when prefill and decode occur.

**Cross-Device and Length Studies.** Base-M2 repeats the three-policy comparison five times at device-local  $B = 1.25$ ; the earlier M2 Pro compares Fixed-1 and PhaseGate seven times at  $B = 1.10$  without TimeGate. Without retuning, M4 uses five repetitions at 64/128/512 output tokens and M2 adds three at 512; devices and lengths are analyzed separately with 10,000 paired-bootstrap resamples.

**Token-Level Measurement.** An in-memory monotonic clock records token and phase times without a monitoring process; we report TPOT, p99 inter-token gaps, p95 per-request maximum gaps, and the phase-transition gap from the end of prefill to the first decoded token.

**Validity Checks.** Evaluation begins only after repeated no-retrieval p95 TPOT and TTFT measurements stay within 3% of their medians; swap growth, abnormal pressure, thermal warnings, corrupt traces, incorrect policy behavior, or severe drift invalidate a run, with one retry allowed. One M4 TimeGate run uses that retry; the other primary runs are first attempts. Small system-wide pageouts under zero swap and normal pressure remain warnings; all 21 M4 runs pass mandatory checks, with one of seven matched three-policy repetitions entirely pageout-free versus all five on M2.

## 5 Results

### 5.1 Phase-Specific Caps Unlock Latency-Safe Throughput

Fixed-1 is the fastest tested constant policy within budget; Fixed-2/4 retrieve more but violate TPOT. PhaseGate 4 $\rightarrow$ 1 keeps decode at one and raises only prefill, reaching 1.6 kQPS versus Fixed-1’s 0.78 on separate requests. Both meet TPOT and TTFT in every run; PHASEGATE therefore doubles *latency-constrained* rather than merely raw throughput.

TimeGate reaches 1.5 kQPS but raises p95 TPOT by 60% and passes no run: cap four remains active for 23% of decode versus 0.0030% with PHASEGATE, so phase-blind time matching fails.

Table 3: Base-M4 evaluation at  $B = 1.25$ . Seven paired randomized repetitions use 250 requests per policy; “7/7” means both latency limits pass in all seven.

| Metric                      | FIXED-1 | PHASEGATE 4 $\rightarrow$ 1 | TimeGate 4/1 |
|-----------------------------|---------|-----------------------------|--------------|
| Runs meeting both limits    | 7/7     | 7/7                         | 0/7          |
| p95 TPOT increase (%)       | 17      | 18                          | 60           |
| p95 TTFT increase (%)       | 0.61    | 4.6                         | 4.3          |
| Retrieval throughput (kQPS) | 0.78    | <b>1.6</b>                  | 1.5          |
| p95 transition gap (ms)     | 18      | 23                          | 24           |

## 5.2 The Effect Reproduces and Shrinks with Output Length

On M4, PhaseGate/Fixed throughput falls from  $2.3\times$  at 64 tokens to  $2.0\times$  at 128 and  $1.4\times$  at 512 as prefill’s share of LLM processing time falls from 69% to 52% and 21% (Figure 2b). Policies are unchanged; the association is descriptive.

Table 4: Mac mini evaluations (two significant digits); safe runs meet both limits, and “–” denotes an unevaluated control.

| Device       | $B$  | PhaseGate/Fixed throughput ( $\times$ ) | PhaseGate/TimeGate throughput ( $\times$ ) | Fixed safe runs | PhaseGate safe runs | TimeGate safe runs |
|--------------|------|-----------------------------------------|--------------------------------------------|-----------------|---------------------|--------------------|
| Base-M4 Mini | 1.25 | 2.0                                     | 1.0                                        | 7/7             | 7/7                 | 0/7                |
| Base-M2 Mini | 1.25 | 1.8                                     | 1.0                                        | 5/5             | 5/5                 | 0/5                |
| M2 Pro Mini  | 1.10 | 2.5                                     | –                                          | 7/7             | 7/7                 | –                  |

With a separate baseline, M2 falls from  $1.8\times$  at 128 tokens to  $1.4\times$  at 512. M2 Pro yields  $2.5\times$ , but lacks TimeGate and pageout-free pairs, supporting only the policy ordering.

## 5.3 Transition and Memory Costs Are Bounded but Visible

Non-preemptive searches may briefly cross into decode, adding 5.2 ms to the p95 phase-transition gap versus Fixed-1—only 0.29% of the 1.8 s baseline TTFT once per request. The p99 inter-token gap and p95 per-request maximum gap rise by less than 1%. All runs have zero swap growth and normal pressure, but system-wide pageouts remain a caveat. Removing any matched comparison, including the one with the largest pageout imbalance, leaves PhaseGate/Fixed at  $2.0\times$ .

## 6 Related Work

**LLM Serving and Interference.** DistServe separates phases across server GPUs, Sarathi-Serve chunks prefill, and speculative decoding reshapes decode [7, 8, 18]. They schedule LLM inference itself; PHASEGATE instead admits CPU retrieval at its phase boundary. BWLOCK++, REEF, and SERENO control traffic, preemption, or observed interference [3, 9, 10]; PHASEGATE anticipates risk from the victim phase.

**On-Device and Agentic Work.** Apple-Silicon profiling, CPU-heavy agents, mobile vector search, and heterogeneous schedulers establish adjacent co-execution problems [4, 11–14, 20]. None compares a fixed retrieval limit with both phase-aligned and time-matched phase-blind controls.

## 7 Limitations and Conclusion

We use one base-M4 Mini for the full scheduler protocol, M2/M2 Pro Minis for narrower replications, and M4/M1 Airs for mechanism tests—one machine each, without pooling. Scheduler evaluations use one 1.5B 4-bit model, a 2,048-token prompt, read-only HNSW, and saturated demand [15, 16]; Air controls add synthetic/native kernels. We omit embedding/index updates; caps are deployment-specific, and bursty demand should reduce gains. Protocol/pageout differences preclude cooling or paging-neutrality claims, and length trends are descriptive. Within these bounds, Fixed-2/4 fail, PHASEGATE doubles latency-safe retrieval throughput over Fixed-1, and TimeGate shows phase-blind clock matching fails. Code, traces, and scripts will be released.

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